

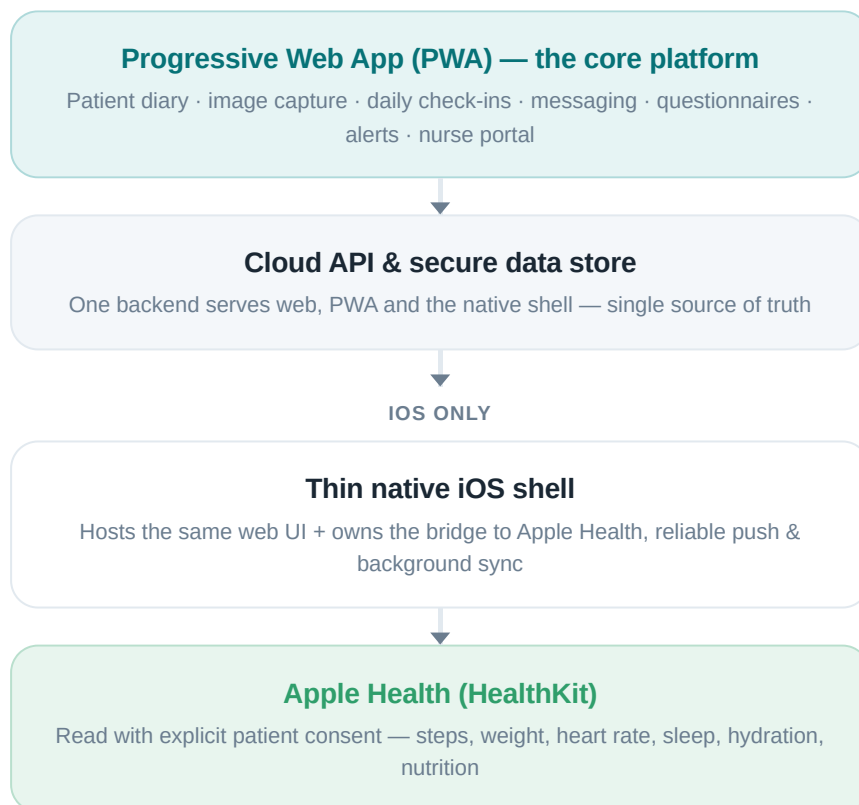


HOW WE BUILD THE MOBILE PRODUCT

PWA core + a thin native iOS shell

The lowest-cost route to a real mobile product that still reaches **Apple Health** and future wearables — without committing to a full native build before the value is proven. This is the standard pattern for NHS-facing digital health (Huma, Ada and peers ship this way).

01 Recommended architecture



One iOS app, not two. The native shell wraps the existing web UI *and* contains the HealthKit module — the patient sees a single icon. Android and desktop users get the plain PWA. This keeps one codebase for all the product features and confines native work to the few things only native can do.

02 Why each piece earns its place

PWA for the core

- One codebase, instant updates, no app-store gate on every change

Native only where it's forced

- **Apple Health is walled off from the web** — a PWA cannot read it. Native is the *only* way in.

- Installable to the home screen from a link — looks like an app
- Carries every feature: diary, check-ins, messaging, education, supplies, outcomes

- On iOS, PWAs have weak push, no real background sync, and can be evicted — the shell fixes all three.
- Future wearable / device integrations land here too.

03 What Apple Health unlocks (with consent)

Steps & activity · weight · heart rate · blood pressure (from connected devices) · sleep · hydration · nutrition · mobility. These can be correlated with the signals that matter most in stoma care — **dehydration risk (especially high-output ileostomy)**, post-surgical recovery trajectory, and overall wellbeing — to surface patients who may need attention sooner.

⚠ The real cost driver isn't the app — it's the regulatory line

The architecture above is cheap and well-trodden. The budget and timeline live in **where you draw the clinical line**. Today's product **displays and organises** data for a nurse to review — non-diagnostic, lower exposure. The moment the product starts to **interpret** data ("these readings suggest dehydration risk"), it likely becomes **Software as a Medical Device**: UKCA / UK MDR, plus NHS **DTAC** and clinical-safety standards **DCB0129 / DCB0160**. Decide which side of that line each feature sits on *before* building it.

04 Sequencing — value first, regulation later

STAGE	BUILD	WHAT IT DELIVERS
Now · shipping	PWA core	Engagement & workflow: diary, check-ins, messaging, education, supplies, outcome tracking, nurse portal. Non-diagnostic.
Next · when HealthKit / wearables are needed	Thin native iOS shell	Apple Health read access, reliable push & background sync. Small, contained native effort.
Later · once adoption is proven	Regulated clinical layer	AI-assisted change detection & decision support, taken through DTAC / DCB0129 / UK MDR.

The installable PWA preview is already live — it **is** stage one of this plan, not a throwaway. Try it on a phone: jsbrown88.github.io/stoma-alert-site/app/

Recommendation. For an NHS-facing product, build a PWA + thin native HealthKit wrapper rather than a fully native app. It minimises development cost while still giving Apple Health access and a path to wearables — and it keeps the regulated clinical layer cleanly separable until the value is proven.

